

Training Configuration and Summary:

Dataset used: rssi_odom_dataset_11apr25_ajust.parquet

model_type: LSTM
group_sec: 1
timesteps_orig: 30
timesteps: 30
batch_size: 256
min_units: 32
max_units: 256
units_step: 32
dropout_min: 0.2
dropout_max: 0.3
dropout_step: 0.1
activation_functions: ['tanh', 'relu', 'elu']
optimizers: ['adam', 'rmsprop', 'Nadam']
min_layers: 1
max_layers: 2
max_epochs: 150
hyperband_iterations: 3
forecast_steps: 2

LSTM Multi-Step Training Report

Report Generated: 2025-05-09 10:01:05

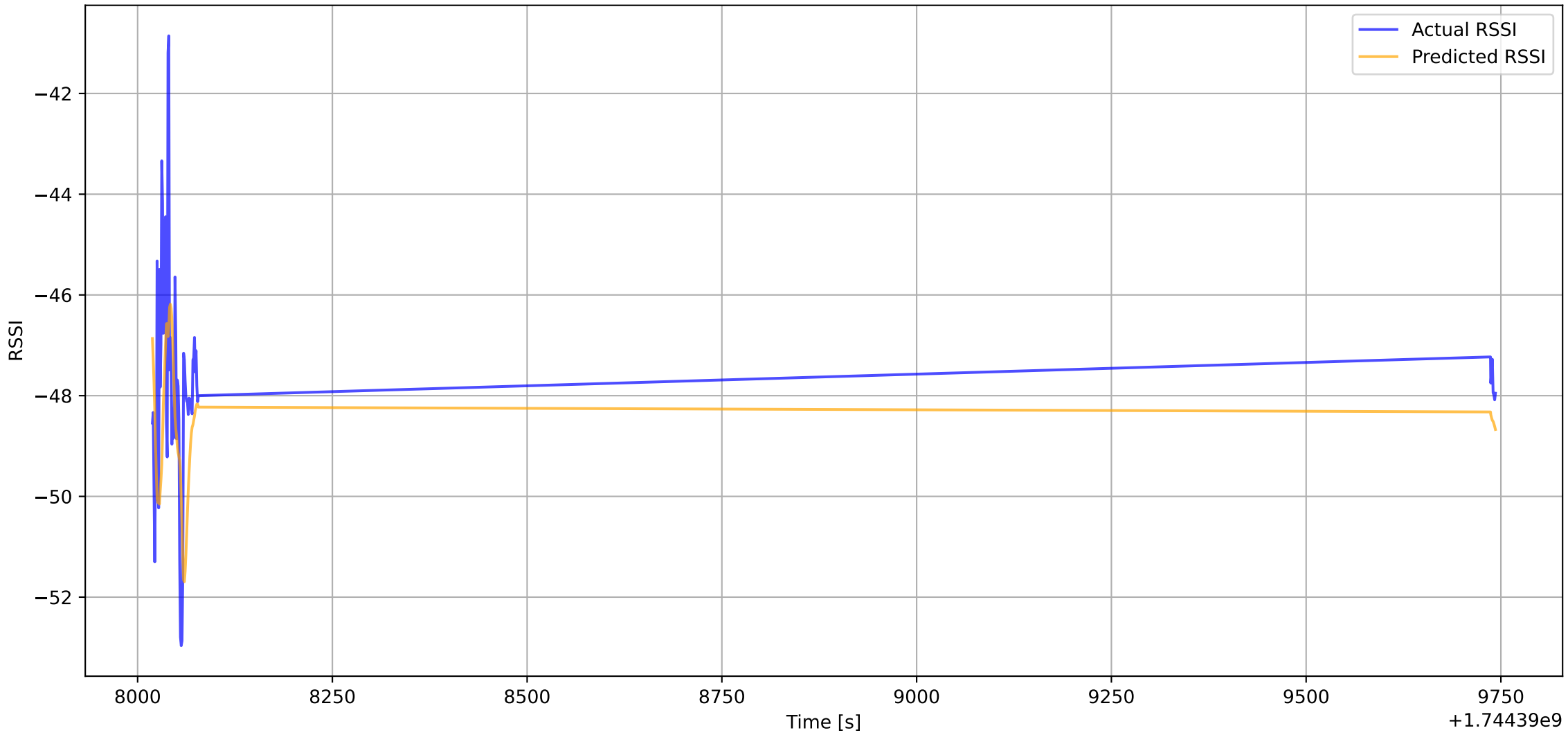
Training Start: 2025-05-09_09-59-14
Training End: 2025-05-09_10-00-40
Training Duration: 1.44 minutes

Best Hyperparameters:
Number of Layers: 1
Activation: elu
Optimizer: Nadam
Dropout Rate: 0.20

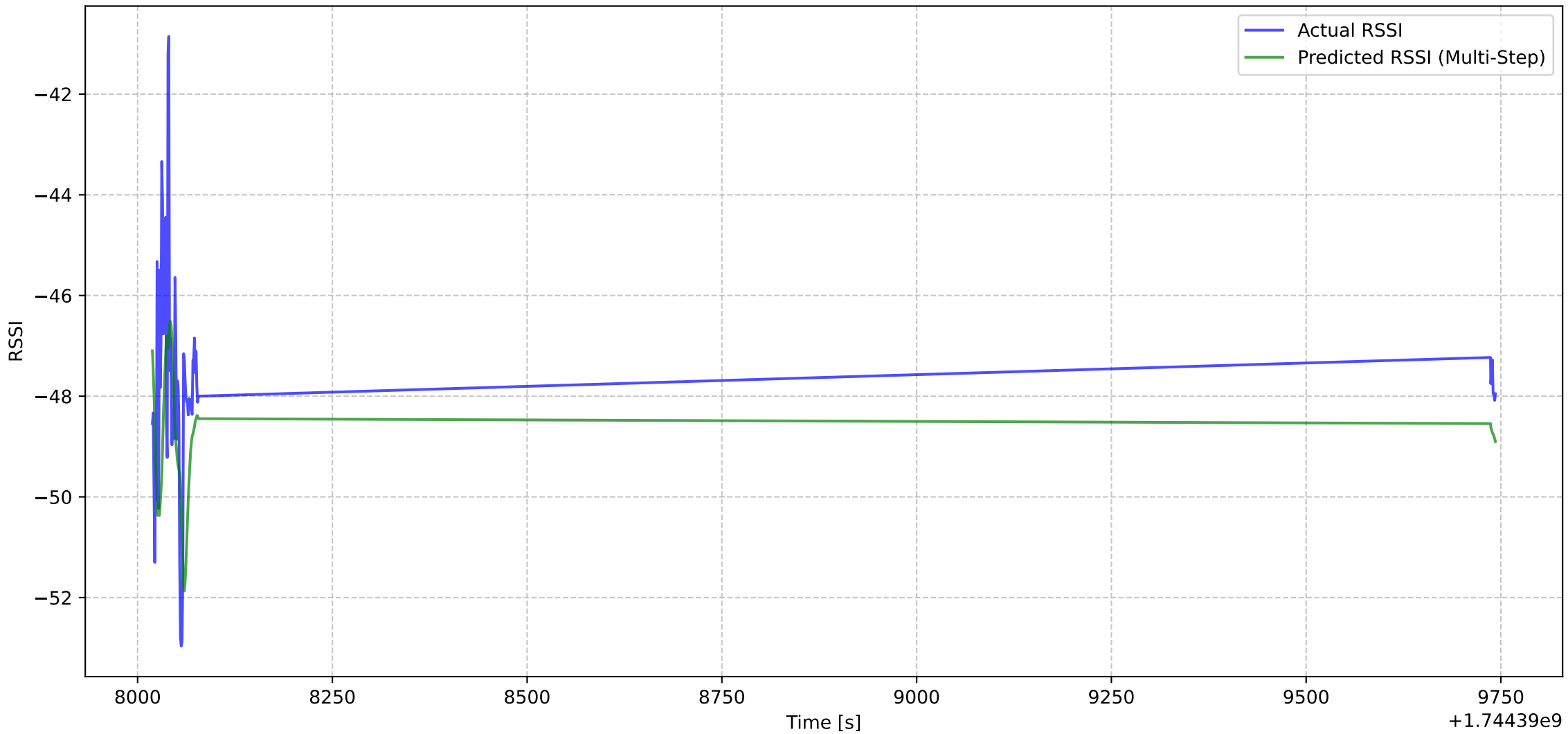
MAE: 1.9870
MSE: 6.1790
RMSE: 2.4858
R2 Score: -0.3156

Multi-Step Forecast:
Forecast Steps: 2
MAE: 0.0817
MSE: 0.0106
RMSE: 0.1027
R2 Score: -0.2134

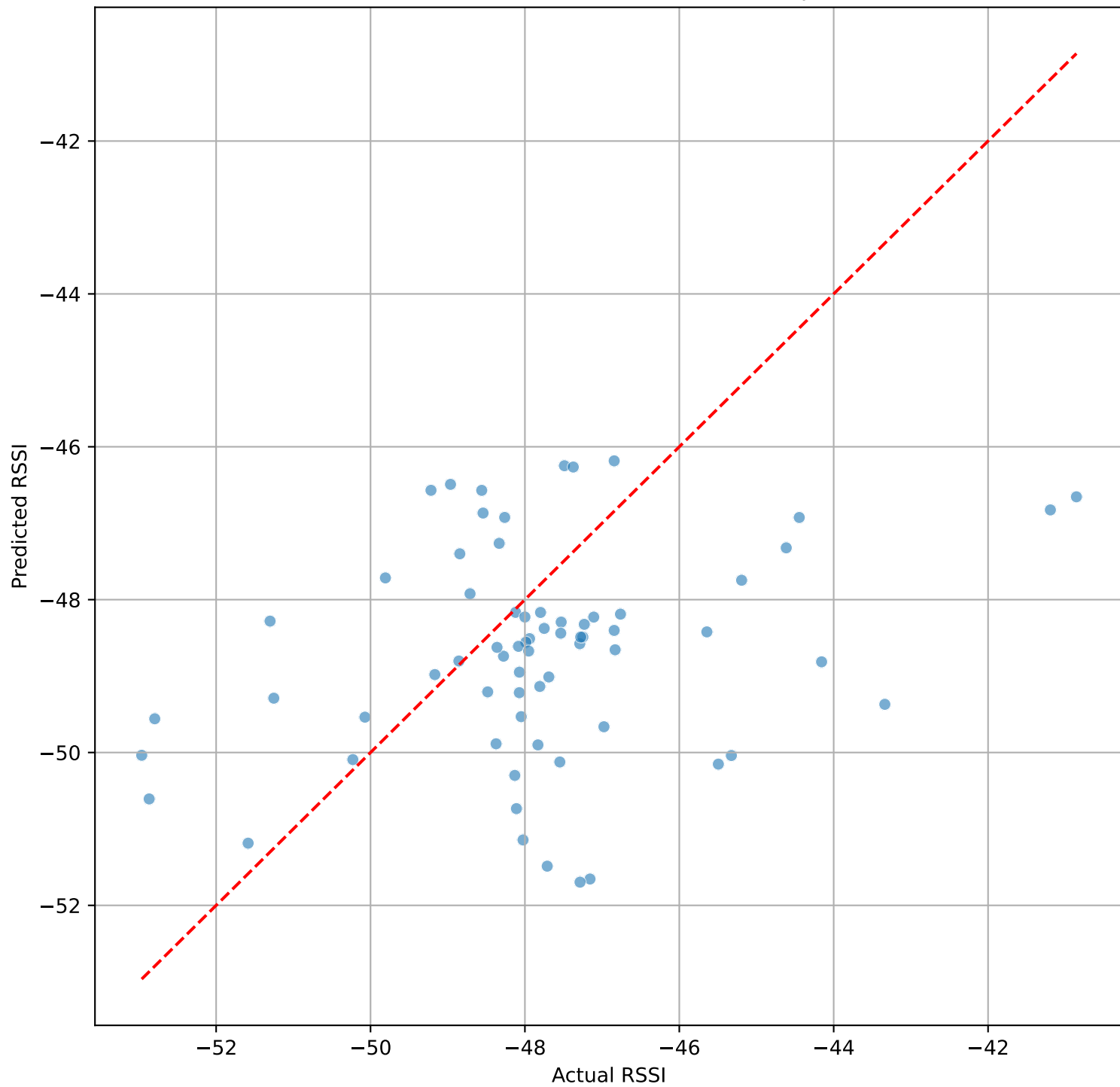
Actual vs Predicted RSSI (First Step)



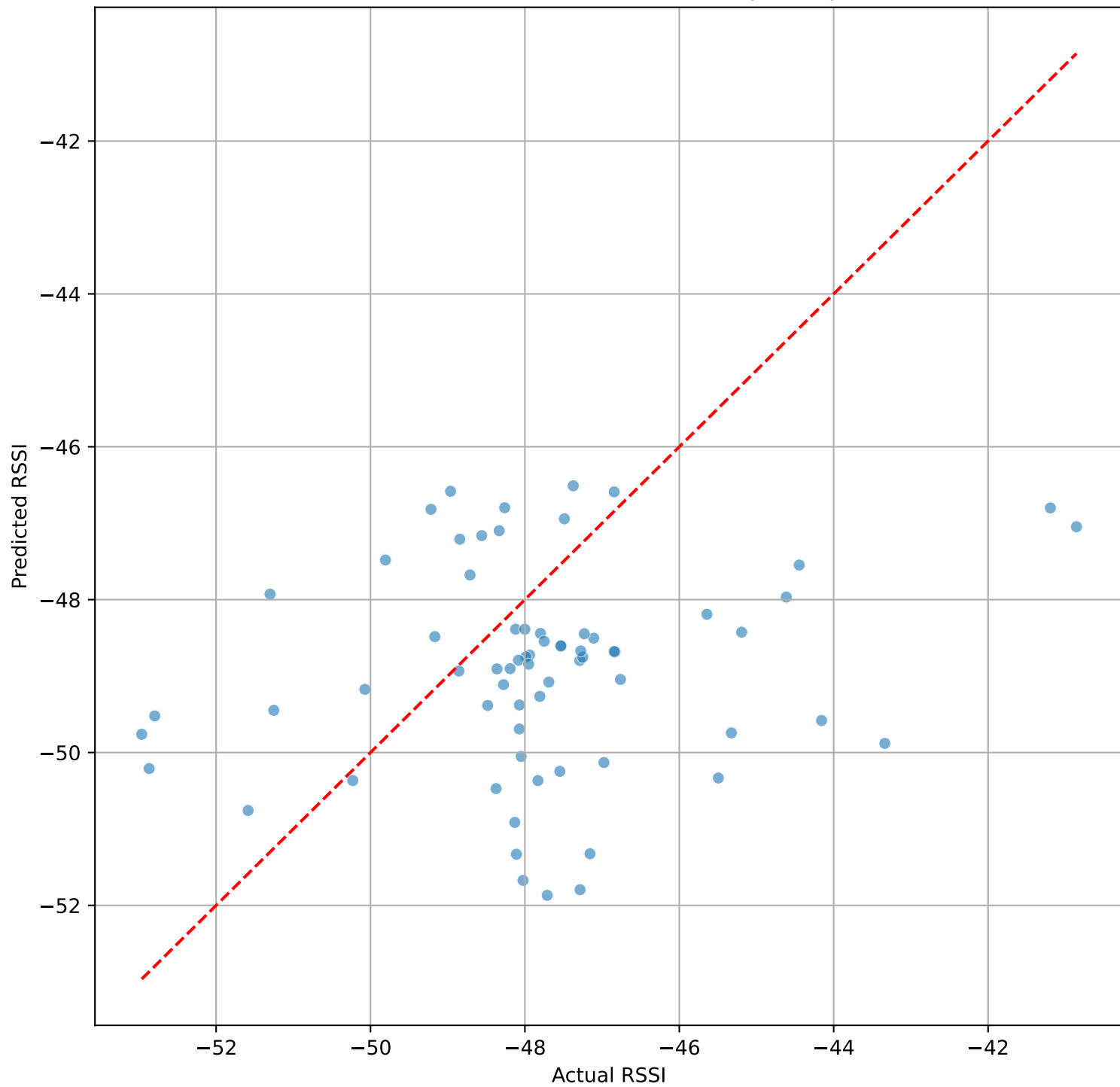
Actual vs Predicted RSSI (Multi-Step (step 2))



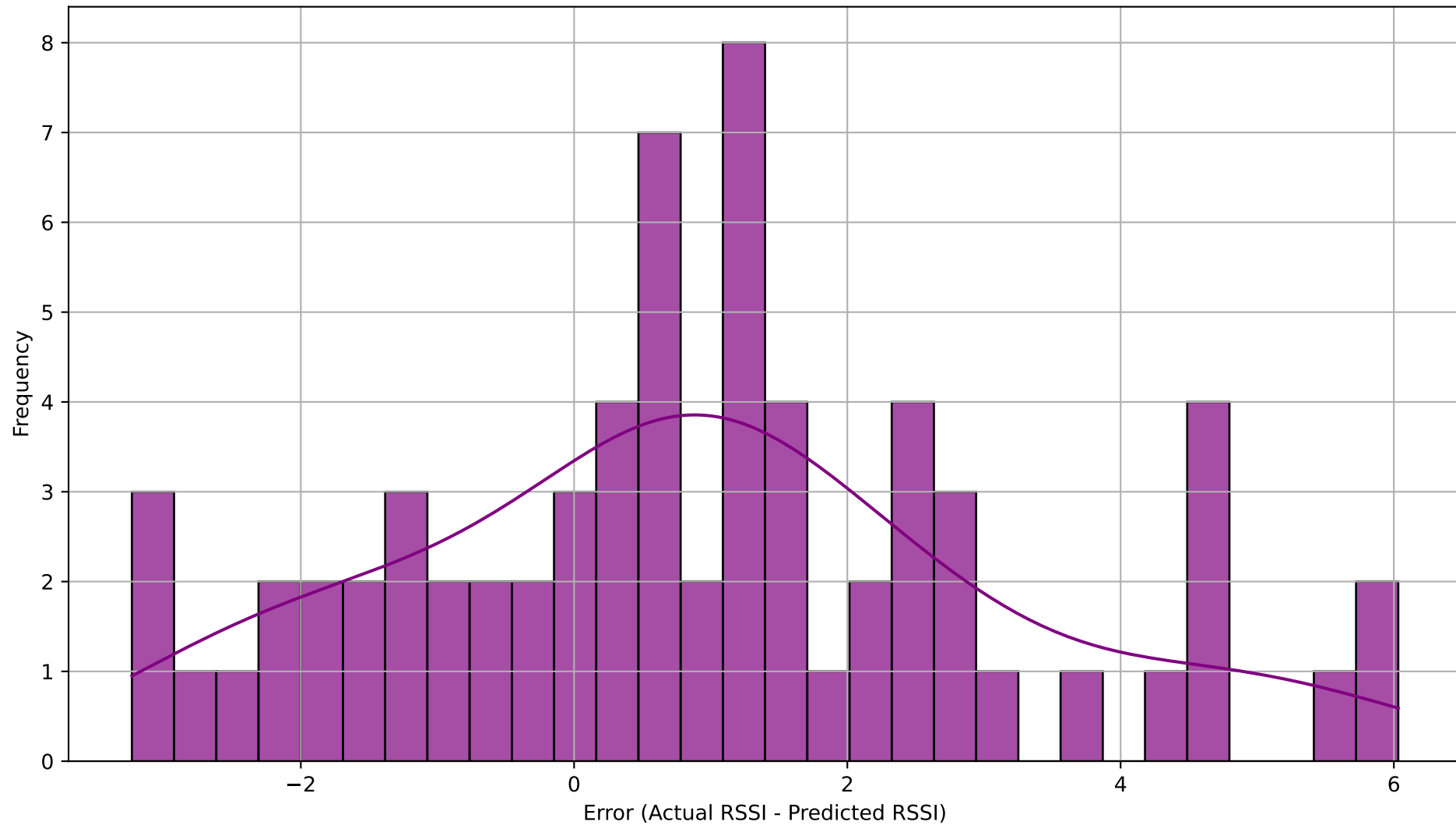
Actual vs Predicted RSSI (First Step)



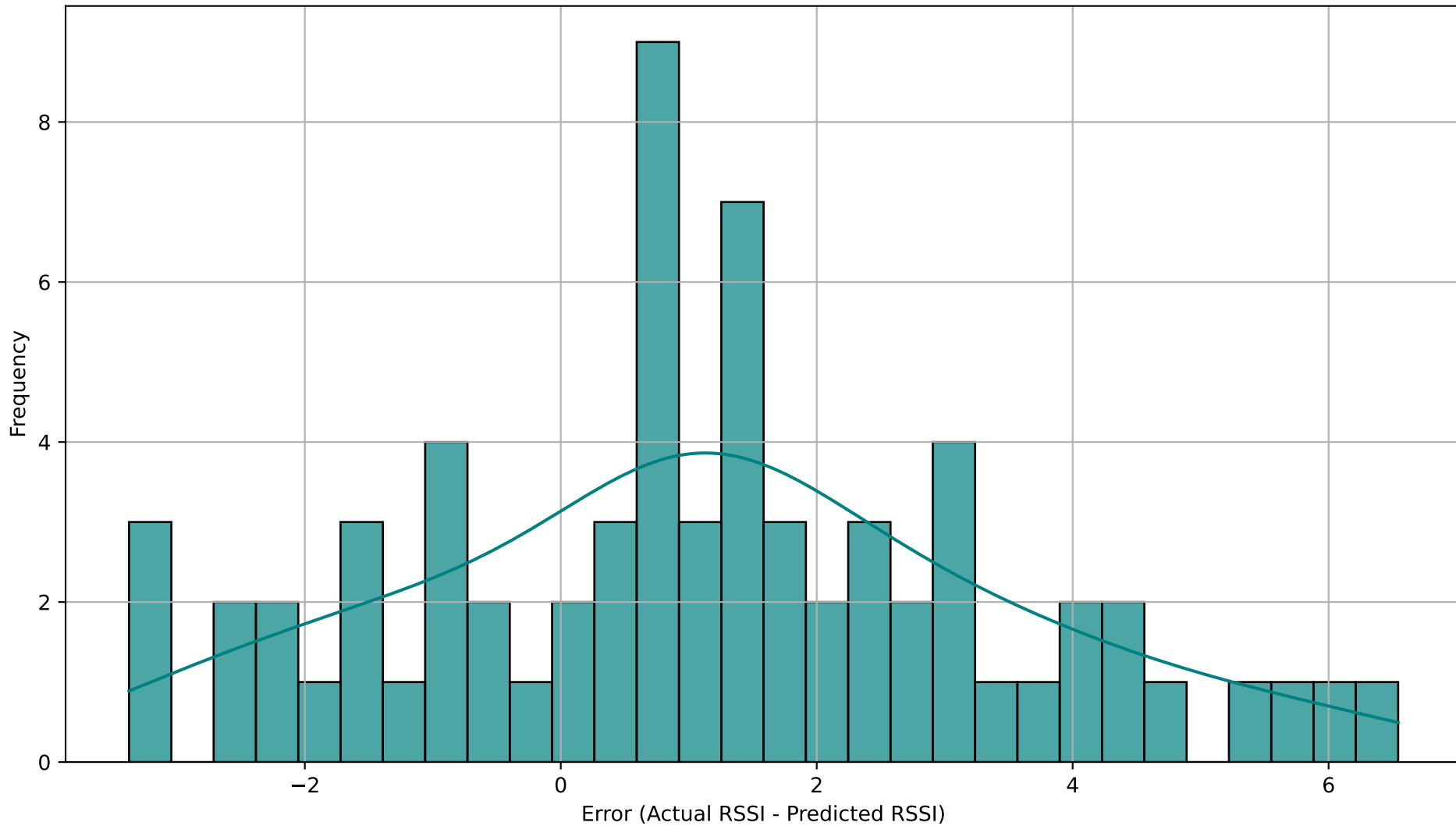
Actual vs Predicted RSSI (Last Step - step 2)



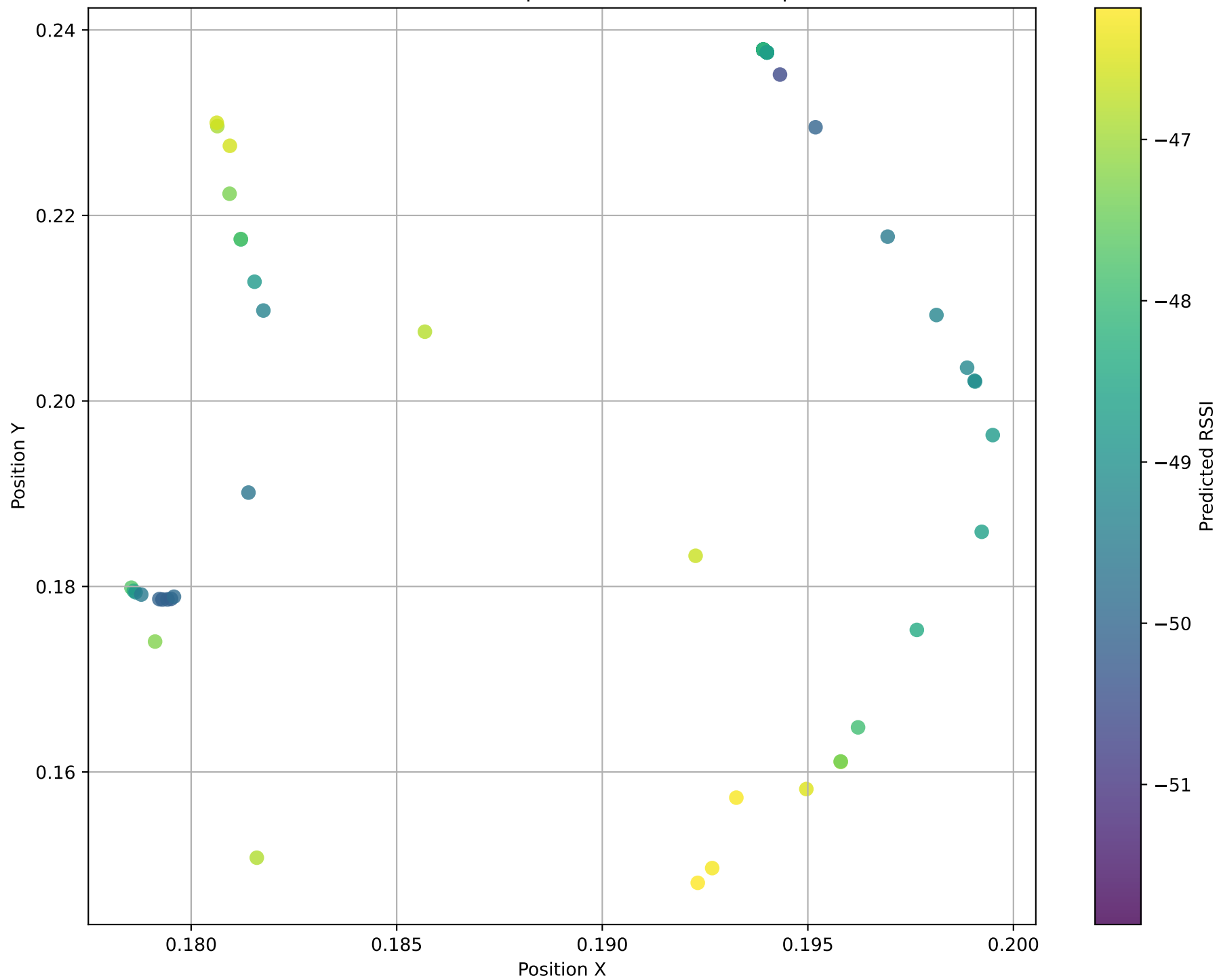
Distribution of Prediction Errors (First Step)



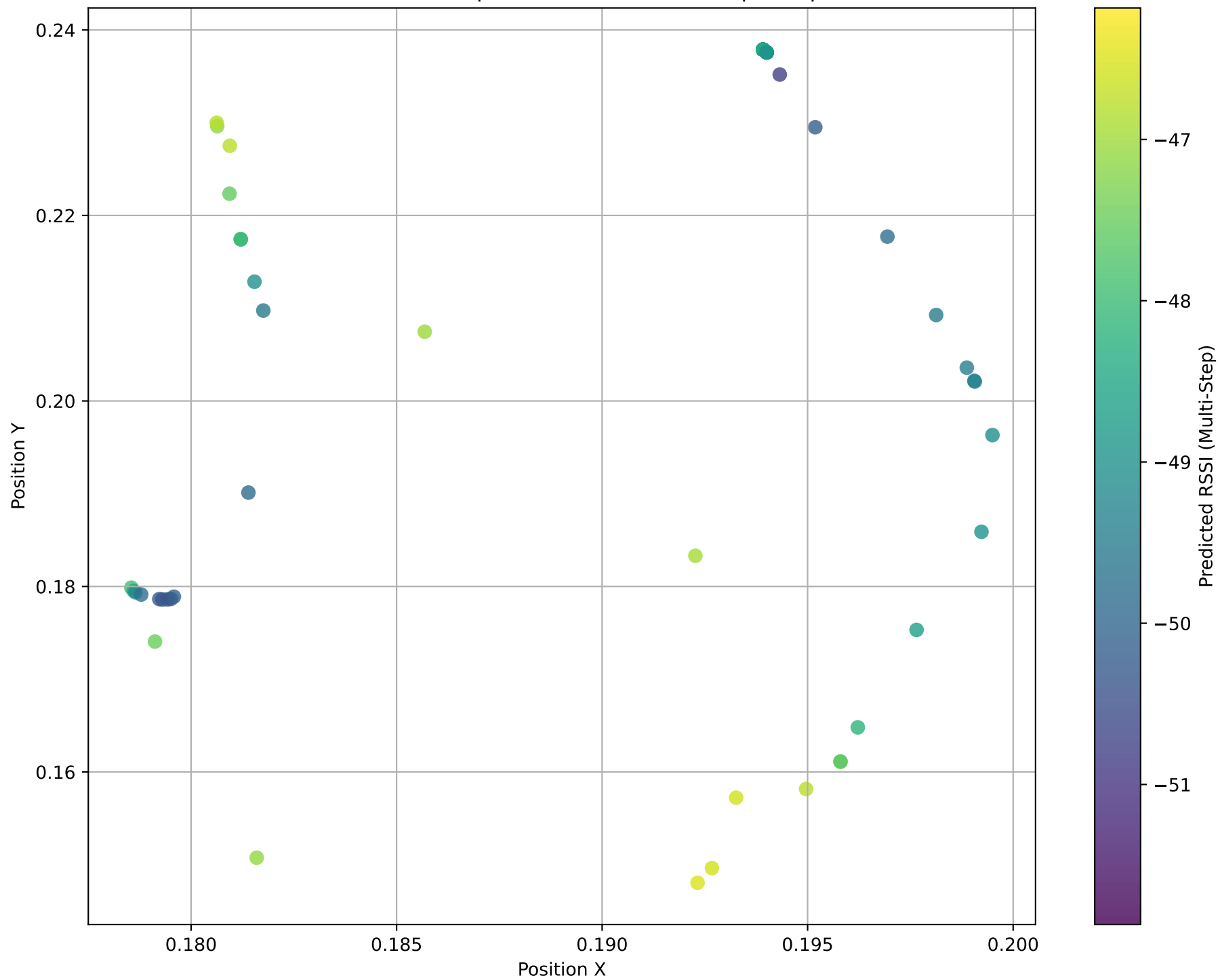
Distribution of Prediction Errors (Last Step - step 2)



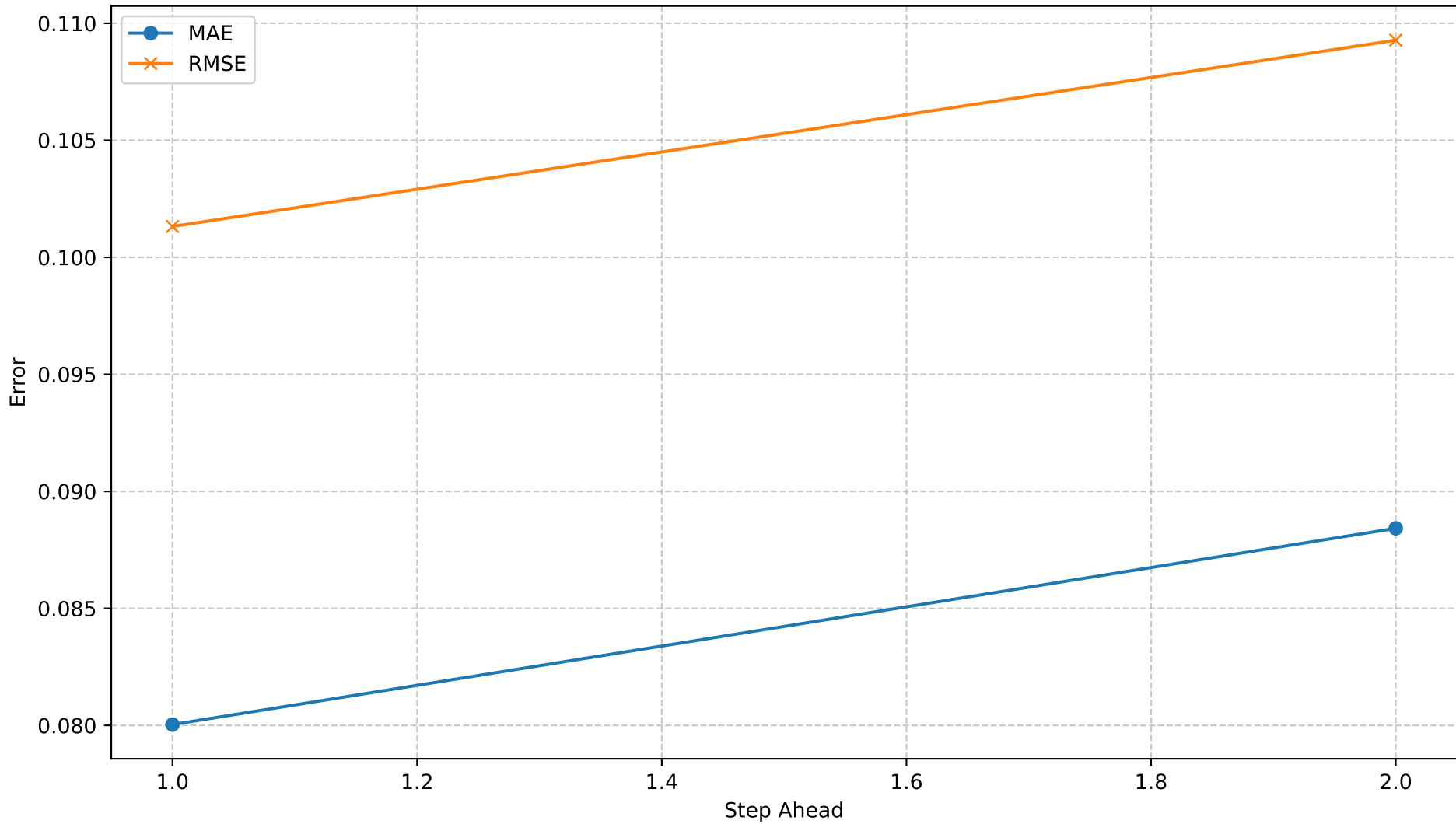
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 2)



Multi-Step Forecast Error by Step



Resumo Automático

Dataset: rssi_odom_dataset_11apr25_ajust.parquet

Modelo: LSTM

Forecast steps: 2

MAE / RMSE (dB)

- 1º passo : 0.08 / 0.10
- Mediana : 0.08 / 0.11
- 2º passo : 0.09 / 0.11

Crescimento do erro: 10.5% → Estável (erro cresce < 20 %)

3 piores passos (MAE): 2 (0.09), 1 (0.08)

Recomendação:

0 modelo é adequado para decisões de hand-over até o horizonte máximo configurado.